MATH 324 Winter 2025 - Tutorial 1

Section 003 (Thursday)

January 16, 2025

- Time and Location: Thursday 1:35-3:25pm in Burnside 1214.
- Office Hours: Wednesday 12-1pm in Burnside 1017
- Email: alicia.ter-cheam@mail.mcgill.ca
- Format: Each tutorial will contain some review of the previous week's material, as well as going over questions that will be posted before tutorial. Written notes of the review material and solutions will be posted after tutorial.

1 Review of Random Variables and Probability Theory

1.1 Random variables

A *random variable* $X : S \to \mathbb{R}$ is a function assigning each sample point $s \in \Omega$ to a real number.

The *cumulative distribution function* (cdf) is

$$F(x) = \mathbb{P}(X \geq x) = \sum_{x_i \leq x} p(x_i).$$

1.1.1 Discrete random variables

A *discrete random variable* is when X can take on discrete values $x_1, x_2, ..., x_k$ with probabilities $P(X = x_1) = p(x_1), p(x_2), ..., p(x_k)$, respectively. We call $p(x)$ the *probability mass function* (pmf), which satisfies:

- $p(x) \geq 0$
- $\sum_{i=1}^{k} p(x_i) = 1$

$x_1, ..., x_k$ are the support of X, or *supp*(X). We can also extend to countably infinite $x_1, x_2, ...$

1.1.2 Continuous random variables

The p.d.f. of a continuous random variable satisfies

- $f(x) \geq 0$ for all x (though now $f(x)$ can be greater than one, unlike the PMF $p(x)$).
- $\int_{-\infty}^{\infty} f(x)dx = 1$

The p.d.f. is the continuous random variable analog to the discrete PMF. Note that for a continuous random variable, for any given x, $\mathbb{P}(X = x) = 0$. The area under the density curve represents the probability of X falling between the corresponding x values.

Thus, the CDF of a continuous random variable is defined by

$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f(t)dt$$

When $F(x)$ is continuous, we say that X is a continuous random variable.

A special continuous distribution: The *Normal distribution* (or Gaussian distribution) with parameters μ and σ^2 is $X \sim N(\mu, \sigma^2)$ with

$$f(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

for $x \in \mathbb{R}$. This is the standard "bell curve" distribution. The parameters are the mean, i.e. $\mathbb{E}[X] = \mu$ and variance, i.e. $\mathrm{Var}(X) = \sigma^2$.

1.2 Expectation and variance

1.2.1 Expectation

Expectation of a discrete random variable X is its mean, with definition:

- $\mathbb{E}[X] \equiv \sum_{x \in Supp(X)} xp(x)$ for a discrete random variable, and
- $\mathbb{E}[X] \equiv \int_{-\infty}^{\infty} xf(x)dx$ for a continuous random variable.

In general, for a function $g : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}[g(X)] \equiv \sum_{x \in Supp(X)} g(x)p(x), \text{ or } \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Useful properties: For $a, b \in \mathbb{R}$:

- Linearity of $\mathbb{E}$: $Exp[aX + b] = a\mathbb{E}[X] + b$
- $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$, i.e. translation does not affect dispersion
- $\mathbb{E}[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} \mathbb{E}[X_i]$
- When X_i's are pairwise independent, $\mathrm{Var}[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} \mathrm{Var}[X_i]$

1.2.2 Variance

The *variance* of a random variable X is defined as

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sigma^2$$

where $\mu = \mathbb{E}[X]$. It can also be shown that

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The *standard deviation* of X is

$$\sigma = \mathrm{Std}(X) = \mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)}.$$

1.3 Multivariate random variables and independence

For random variables X and Y, the random vector (X, Y) has joint CDF

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y).$$

X and Y are *independent*, written $X \perp\!\!\!\perp Y$, if and only if

$$F_{X,Y}(x, y) = F_X(x)F_Y(y).$$

Equivalently, we have independence if

- X, Y are discrete: $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$
- X, Y are continuous: $f_{X,Y}(x, y) = f_X(x)f_Y(y)$

1.4 Central limit theorem

If $X_1, X_2, ..., X_n$ are i.i.d. random variables with common expectation μ and variance σ^2, then by CLT, the sample mean $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$.

Question 1 - Central limit theorem

To understand central limit theorem, we will be generating samples of varying sizes and exploring the behaviours of the sample means and variances.

(a) Generate 1000 samples of 10 Poisson($\lambda = 1$) observations, saving the sample mean and variances. Plot the histogram of both.

(b) Now, generate 1000 samples of 100 Poisson($\lambda = 1$) observations, saving the sample mean and variances. Once again, plot the histogram of both. Contrast with plots from **(a)**.

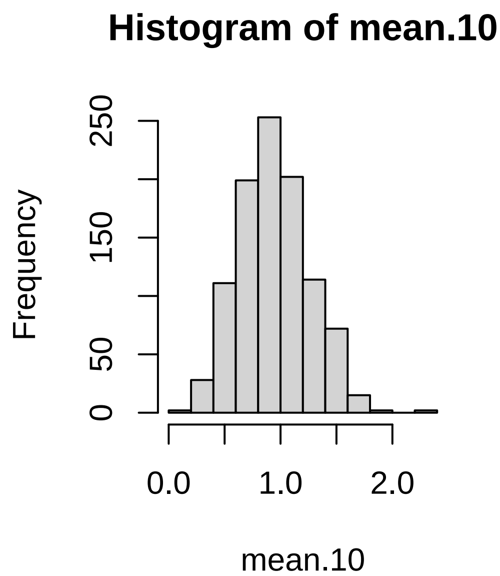
(c) In these two examples, we are considering $X_i \sim$ Poisson(1) for $i = 1, ..., n$. In the first case, $n = 10$ and then in the second, $n = 100$. Find the general distribution for $\bar{X}_n = \sum_{i=1}^{n} X_i$.

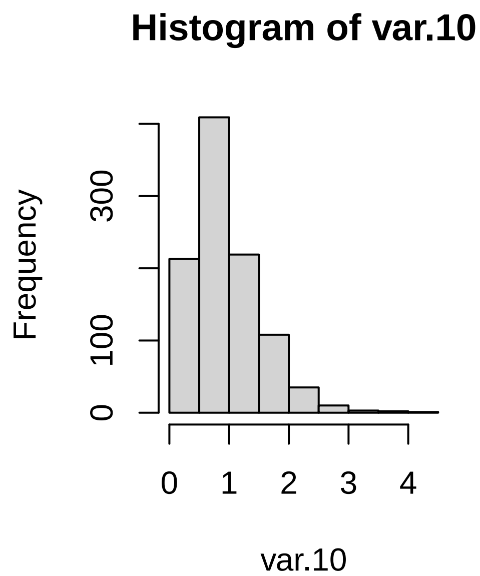
(d) Explore the variance of the sample means generated in **a** and **b**. How does this relate to the distribution of $\bar{X}_n$?

Solution

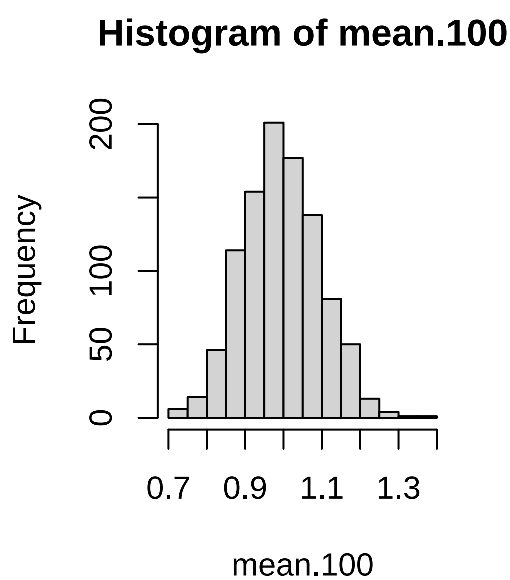
(a)

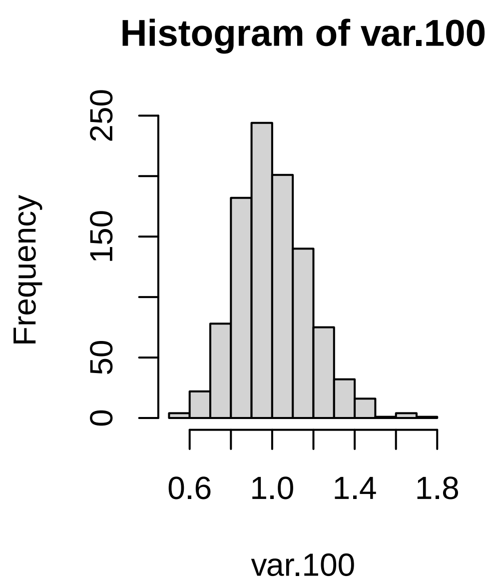
```
mean.10 <- rep(0,1000)
var.10 <- rep(0,1000)

for(i in 1:1000){
    sample <- rpois(10,1)
    mean.10[i] <- mean(sample)
    var.10[i] <- var(sample)
}

par(mfrow=c(1,2))
hist(mean.10)
hist(var.10)
```





(b) Code is omitted for brevity; we just change `rpois(10,1)` to `rpois(100,1)` when generating the sample. We obtain the following plots:





(c) $X_i \sim \text{Pois}(\lambda_i)$. As the X_i are independent, then $Y = \sum_{i=1}^n X_i \sim \text{Pois}(\sum_{i=1}^n \lambda_i) = \text{Pois}(n\lambda)$ as $\lambda_i = \lambda$ for all i.

Then,

$$\mathbb{E}[\bar{X}_n]\mathbb{E}\left[\frac{1}{n}Y\right] = \frac{1}{n}\mathbb{E}[Y] = \frac{1}{n}(n\lambda) = \lambda$$

and

$$\text{Var}\left(\frac{1}{n}Y\right) = \frac{1}{n^2}\text{Var}(Y) = \frac{1}{n^2}(n\lambda) = \frac{\lambda}{n}$$

Recall that by CLT, with n large, then $\bar{X}_n \approx N(\lambda, \frac{\lambda}{n})$.

Note: We show how we determine the distribution for the sum of Poisson distributions using the MGF. Recall that the MGF of the Poisson is $M(t) = e^{\lambda(e^t-1)}$. Then if $Y = \sum_{i=1}^n X_i$,

$$M_Y(t) = \mathbb{E}[e^{yt}] = \mathbb{E}\left[\prod_{i=1}^n e^{x_i t}\right] = \prod_{i=1}^n \mathbb{E}[e^{x_i t}] = (M_X(t))^n = e^{n\lambda(e^t-1)}$$

which is the MGF of Pois($n\lambda$). We require independence to pull the product out of the expectation and use the fact that they are i.i.d.

Question 2 - Sensitivity of the Sampling Distribution

Suppose we have a population that is Normally distributed with mean 71 and variance 55.

(a) If we take a sample size of 8, what is the smallest sample mean needed for us to say we are 99% confident that the sample has a larger mean?

(b) Now suppose we take a sample of size 200. How little does the sample mean have to differ from our true mean for it to fall outside the 95% confidence interval?

Solution

(a) $X_i \sim N(71, \sigma^2 = 55)$ and $\bar{X}_n \sim N(\mu, \sigma^2 = \frac{55}{8})$, with $n = 8$.

If we are 99% confident, there is at most a 1% chance that the true mean would be rejected from a sample constructed using $\bar{x}$. Using the variance of $\bar{X}_n$, we would construct a one-sided confidence interval of (L, ∞). L is determined as

$$\mathbb{P}(\bar{X}_n \leq L) \leq 0.01.$$

Note that as $\bar{X}_n$ is assumed to share the same variance as the true population X, this is equivalent to finding x such that

$$\mathbb{P}(X \geq x) \leq 0.01$$

. where X has mean 71.

```
qnorm(0.01, mean=71, sd=sqrt(55/8), lower.tail=FALSE)

## [1] 77.09974
```

Thus, $x = L = 77.1$

Note: Originally, the question asked for the smallest sample mean so we are confident the sample has a *different* mean. The generality of a 'different' mean leads us to want to construct a two-sided interval and thus smallest sample mean is ambiguous. The wording has been changed to be clearer in what is needed.

(b) $\bar{X}_n \sim N(71, \sigma^2 = \frac{55}{200})$. We can directly calculate an approximate 95% confidence interval as $\mu_{\bar{X}} \pm 2\sigma = 71 \pm 2\sqrt{55/200}$ which is approximately 1.05.

2 Estimation, Bias, and MSE

We consider $\hat{\theta}$ to be an *estimator* of an unknown parameter θ. For example, $\hat{\theta} = \bar{X}_n$ is an estimator of $\theta = \mu = \mathbb{E}[X]$.

Our estimator is itself a random random variable as it depends on the random sample it comes from. A *point estimator* is if we assign a single value to $\hat{\theta}$ (e.g. $\bar{X}_n$). There are also *interval estimators*, e.g. confidence intervals.

2.1 Bias

If $\hat{\theta}$ is a point estimator for θ, then $\hat{\theta}$ is *unbiased* if $\mathbb{E}[\hat{\theta}] = \theta$. For example, $\mathbb{E}[\bar{X}_n] = \mu$ so is unbiased for μ.

Question 3 - Sufficiency

Let $\hat{\theta}$ be an estimator for an unknown parameter θ. Define $B(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. Show that

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + (B(\hat{\theta}))^2.$$

Solution

$$\begin{aligned}
\mathbb{E}[(\hat{\theta} - \theta)^2] &= \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}] + \mathbb{E}[\hat{\theta}] - \theta)^2] \quad \text{where } \mathbb{E}[\hat{\theta}] - \theta = B(\hat{\theta}) \\
&= \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}] + B)^2 \quad \text{we will let } B = B(\hat{\theta}) \text{ for shorthand} \\
&= \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2] + \mathbb{E}[B^2] + 2B\mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]] \\
&= \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2] + \mathbb{E}[B^2] + 2B(\mathbb{E}[\hat{\theta}] - \mathbb{E}[\mathbb{E}(\hat{\theta})] \\
&= Var(\hat{\theta}) + B^2 + 2B(\theta - \theta) \\
&= MSE(\hat{\theta})
\end{aligned}$$

Question 4 - Bias

(Exercise 8.3) Let $\hat{\theta}$ be an estimator for an unknown parameter θ and $\mathbb{E}[\hat{\theta}] = a\theta + b$ for some nonzero constants a and b. Find a function of $\hat{\theta}$ that is an unbiased estimator for θ.

Solution

$$\begin{aligned}
\mathbb{E}[\hat{\theta}] &= a\theta + b \\
\mathbb{E}[\hat{\theta} - b] &= a\theta \\
\frac{1}{a}\mathbb{E}[\hat{\theta} - b] &= \theta \\
\mathbb{E}\left[\frac{\hat{\theta} - b}{a}\right] &= \theta
\end{aligned}$$

If we use $\frac{\hat{\theta}-b}{a}$, this estimator is unbiased for θ.

Question 5 - Bias

(Exercise 8.8) Suppose that Y_1, Y_2, Y_3 denote a random sample from an exponential distribution with p.d.f.

$$f(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta}, & \text{if } y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Consider the following five estimators of θ:

$$\hat{\theta}_1 = Y_1, \qquad \hat{\theta}_2 = \frac{Y_1 + Y_2}{2}, \qquad \hat{\theta}_3 = \frac{Y_1 + 2Y_2}{3}, \qquad \hat{\theta}_4 = \min(Y_1, Y_2, Y_3)$$

(a) Which of these estimators are unbiased? (*Hint:* From Exercise 6.81, $\hat{\theta}_4$ has an exponential distribution.)

(b) Among the unbiased estimators, which has the smallest variance?

Solution

Note that $Y \sim \text{Exp}(\theta)$, thus $\mathbb{E}[Y] = \theta$ and $\text{Var}(Y) = \theta^2$.

(a)

$$\begin{aligned}
&\mathbb{E}[\hat{\theta}_1] = \mathbb{E}[Y_1] = \theta \\
&\mathbb{E}[\hat{\theta}_2] = \mathbb{E}\left[\frac{Y_1 + Y_2}{2}\right] = \frac{1}{2}(\theta + \theta) = \theta \\
&\mathbb{E}[\hat{\theta}_3] = \theta \\
&\mathbb{E}[\hat{\theta}_4] = \frac{\theta}{3} \\
&\therefore \hat{\theta}_4 \text{ is biased}
\end{aligned}$$

<u>*Note:*</u> To show that $\hat{\theta}_4$ is exponential, you can directly compute $\mathbb{P}(\hat{\theta}_4 \geq y) = \mathbb{P}(Y_1, Y_2, Y_3 \geq y)$ and use independence to recover the exponential distribution.

(b)

$$\begin{aligned}
\text{Var}(\hat{\theta}_1) &= \theta^2 \\
\text{Var}(\hat{\theta}_2) &= \text{Var}\left(\frac{Y_1 + Y_2}{2}\right) \\
&= \frac{1}{4}\text{Var}(Y_1 + Y_2) \\
&= \frac{1}{4}\left(\text{Var}(Y_1) + \text{Var}(Y_2)\right) \quad \text{by independence (random sample)} \\
&= \frac{\theta^2}{2} \\
\text{Var}(\hat{\theta}_3) &= \frac{1}{9}\left(\text{Var}(Y_1) + \text{Var}(2Y_2)\right) \\
&= \frac{5\theta^2}{9}
\end{aligned}$$